

Francesco Morri

francesco.morri98@gmail.com | francescomorri.github.io | linkedin.com/francesco-morri/
github.com/FrancescoMorri

Education

University of Lille- INRIA Lille • PhD student, thesis subject: <i>Game-Theoretic Learning in Intelligent Marketplaces</i>	Oct 2022 - Dec 2025
Sorbonne Université • M2 in Physics of Complex Systems (i-PCS) • Final Grade: 16/20 (très bien)	Sep 2021 – Jul 2022
Abdus Salam International Centre for Theoretical Physics (ICTP) • Spring College in the Physics of Complex Systems	Feb 2022 – Mar 2022
Politecnico di Torino • MSc in Physics of Complex Systems (International Track) • Final Grade: 110/110 cum laude	Sep 2020 – Oct 2022
Università di Bologna • BSc in Physics • Final Grade: 110/110	Oct 2017 – Sep 2020

Experience

Visiting Student , Universidad de O'Higgins, Rancagua, Chile Supervisor: David Salas I worked on non-cooperative games and conjectural games.	May 2025
Visiting Student , Polytechnique Montréal, Montréal, Canada Supervisors: Quentin Cappart, Hanane Dagdougui I worked on machine learning to enhance optimization algorithms.	Jul 2024 - Sep 2024
Visiting Student , Polytechnique Montréal, Montréal, Canada Supervisor: Quentin Cappart I worked with Quentin Cappart on optimization algorithms for smart building, focusing on the NeurIPS Citylearn Challenge 2023 (which we won). The visit was part of the Galangal project, a collaboration of researchers from Lille, Montréal and Edinburgh.	Nov 2023
Research Intern , Institut de Physique Théorique (IPHT), Saclay, France Supervisor: Pierfrancesco Urbani I studied simple algorithms to solve continuous constraints satisfaction problems close to their satisfiability transition, using statistical mechanics and spin glasses theory.	Mar 2022 - Jul 2022
Visiting Student , SISSA & ICTP, Trieste, Italy I attended courses with PhD students of both SISSA and ICTP as part of my first semester in the Master in Physics of Complex Systems (International Track)	Sep 2020 - Jan 2021

Publications

Winning the 2023 CityLearn Challenge: A Community-Based Hierarchical Energy Systems Coordination Algorithm Authors: A. I. Garmendia, F. Morri, Q. Cappart, H. Le Cadre 27th European Conference on Artificial Intelligence (ECAI)	2024
Learning in Stackelberg Games with Application to Strategic Bidding in the Electricity Market Authors: F. Morri, H. Le Cadre, P. Gruet, L. Brotcorne	2024

20th International Conference on the European Energy Market (EEM)

On the Thermodynamic Interpretation of Deep Learning System

2021

Authors: R. Fioresi, F. Faglioni, F. Morri, L. Squadrani

Geometric Science of Information. GSI 2021. Lecture Notes in Computer Science, vol 12829. Springer

Pre-prints

Nonconvex Game and Multi Agent Reinforcement Learning for Zonal Ancillary Markets

2025

Authors: F. Morri, H. Le Cadre, P. Gruet, L. Brotcorne

Submitted to *Transactions on Energy Markets, Policy and Regulation* in October 2025

Learning in Conjectural Stackelberg Games

2025

Authors: F. Morri, H. Le Cadre, L. Brotcorne

Presented at *Games, Agents, and Incentives Workshop* at AAMAS 2025

Talks and Presentations

GAIW: Learning in Conjectural Stackelberg Games - *F. Morri, H. Le Cadre, L. Brotcorne*

2025 - Detroit, USA

7th Games, Agents, and Incentives Workshop

ROADEF: Learning in Conjectural Stackelberg Games - *F. Morri, H. Le Cadre, L. Brotcorne*

2025 - Paris, France

26ème Congrès Annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision

ISMP: Learning in Multi-Leader Single-Follower Stackelberg Games - *F. Morri, H. Le Cadre, L. Brotcorne*

2025 - Montréal, Canada

25th International Symposium on Mathematical Programming

IMACS: Multi-Agent Reinforcement Learning for Strategic Bidding in the Electricity Market - *F. Morri, H. Le Cadre, P. Gruet, L. Brotcorne*

2023 - Roma, Italy

21st International Association for Mathematics and Computers in Simulation World Congress

Fime Summer School on Big Data & Finance: Multi-Agent Reinforcement Learning for Strategic Bidding in Two Stage Electricity Markets - *F. Morri, H. Le Cadre, P. Gruet, L. Brotcorne*

2023 - Aussois, France

LION17: Multi-Agent Reinforcement Learning for Strategic Bidding in Two Stage Electricity Markets - *F. Morri, H. Le Cadre, P. Gruet, L. Brotcorne*

2023 - Nice, France

17th Learning and Intelligent Optimization Conference

Teaching

Object Oriented Programming, *École Centrale de Lille, G1/G2*

2025

The goal of this course is to introduce the basic concepts of object oriented programming with Java. It is organized in 40h of lab sessions, where the students have to develop small projects following the teacher's instructions.

Contribution: I followed a class of 16 students, teaching the basics of the Java language and helping with coding exercise. I then participated in the oral exam at the end of the course.

Reviewing

Reviewer for IEEE Transactions on Automatic Control

2026

Reviewer for AAAI26

2025

Reviewer for LION 2025

2025

Technical Skills

Coding:

- **Experienced:** C++ (ROOT, GSL), Python (PyTorch, Numpy, Pandas, Matplotlib)
- **Familiar:** JavaScript
- **Basics:** Java, HTML, CSS

Languages:

- **Native:** Italian
- **Fluent:** English
- **Intermediate:** French